

IoT Financial Performance Dashboard

Executive Analytics

Python + Excel + MySQL + Power BI

8,900 Transactions · \$4.67B Revenue · 48.01% Avg Net Margin · 100 Branches · 5 Regions

TOOLS	SOURCE	RECORDS	DIMENSIONS
Python+Excel+MySQL+PBI	IoT Dataset (Kaggle)	8,900 · 46 Cols	5 Regions · 6 Depts

01 EXECUTIVE SUMMARY

- Analyzed an IoT Financial Management dataset of 8,900 transaction records across 100 branches, 5 regions, and 6 departments through a complete 4-tool Advanced pipeline — Python EDA confirmed zero nulls, zero duplicates, and 6 negative sensor values (fixed with `abs()`); Python engineered 7 new analytical columns (Net_Profit_Margin_Pct, Expense_Ratio_Pct, Gross_Profit, EBITDA_Margin_Pct, Profit_Category, Risk_Level, Is_High_Performer) producing a 46-column clean CSV.
- Excel analysis built a Branch KPI comparison PivotTable with conditional formatting (color scales for margins, data bars for ROI, traffic lights for performance), Financial Status cross-tab by Region and Department, and a KPI Summary sheet with formula-driven aggregations — providing an offline analytical layer complementing the MySQL/Power BI pipeline.
- Imported 8,900 clean records into MySQL (financial_project database), created 6 performance indexes, and executed 7 analytical SQL queries plus 1 analytical VIEW (`vw_financial_summary`) — surfacing \$4.67B total revenue, 48.01% average net margin, Branch 1025 as top performer (64.33% margin), East region as highest-margin region (49.20%), and Investment department as highest-margin department (50.49%).
- Built a 3-page Power BI dashboard (24 visuals, 12 DAX measures) with Sync Slicers — Page 1 (Financial Overview with 6 KPI cards, waterfall chart, regional bar, status donut), Page 2 (Department & Profitability with margin bars, revenue vs profit column, profit category, Region×Department matrix heatmap), Page 3 (Branch Performance & Risk with top/bottom branch charts, ROI by region, compliance vs performance scatter).

02 PROBLEM STATEMENT

An IoT-enabled financial management system generated 8,900 transaction records across 100 branches, 5 regions, and 6 departments — capturing revenue, profit, operating costs, EBITDA, risk scores, compliance metrics, and operational KPIs in a single flat file with no pre-calculated financial ratios, no branch performance rankings, no risk-profitability cross-analysis, and no executive reporting layer. Management had no visibility into which branches, regions, or departments were underperforming, which risk profiles correlated with high margins, or how the 77.07% 'Excellent' financial status classification translated into concrete performance differentiation across the \$4.67B revenue portfolio.

03 OBJECTIVES

- Obj 1: Perform Python EDA on 8,900-row, 39-column dataset — fix 6 negative sensor values and engineer 7 derived financial columns.

- Obj 2: Build Excel analytical model with PivotTable branch comparison, conditional formatting KPI thresholds, and financial status cross-tabulation.
- Obj 3: Import clean 46-column dataset into MySQL, create 6 indexes, execute 7 queries covering KPIs, status distribution, regional/department performance, branch ranking, and risk-profitability cross-analysis.
- Obj 4: Build 3-page Power BI dashboard with 12 DAX measures and 24 visuals including waterfall chart, Region×Department matrix heatmap, and compliance vs performance scatter.
- Obj 5: Rank all 100 branches by net profit margin to identify top and bottom performers for management intervention.
- Obj 6: Cross-analyze profit category against risk level to determine whether higher-margin transactions carry disproportionate fraud/credit risk.

04 DATASET OVERVIEW & EDA FINDINGS

Attribute	Detail
Dataset Name	IoT Financial Management Dataset
Source	Kaggle — IoT Financial Management
File Name	IoT_Financial_Management_Dataset.csv
Raw Records	8,900 rows · 39 columns
Final Records	8,900 rows · 46 columns (7 derived by Python)
Null Values	Zero ■
Duplicates	Zero ■
Negative Values Fixed	6 rows: 3 in Network_Latency_ms + 3 in Transaction_Processing_Time_ms → abs()
Zero Values Kept	Fraud/Anomaly/Credit scores with zero = valid (very low risk readings)
Dimensions	5 Regions · 6 Departments · 100 Branch IDs · 8,900 Device IDs
Financial_Status	Excellent 77.07% · Good 13.20% · Moderate 6.82% · Poor 2.91%
No Date Column	Dataset has no Year/Date → no YoY or CAGR analysis — used Region/Branch as time-equivalent dimension

Python — 7 Derived Columns Created

Column	Formula	Business Purpose
Net_Profit_Margin_Pct	$\text{Net_Profit} / \text{Revenue} \times 100$	Primary profitability KPI — mean 48.01%
Expense_Ratio_Pct	$\text{Operating_Cost} / \text{Revenue} \times 100$	Cost efficiency metric — mean 81.02%
Gross_Profit	$\text{Revenue} - \text{Operating_Cost}$	Absolute gross profit in \$ terms
EBITDA_Margin_Pct	$\text{EBITDA} / \text{Revenue} \times 100$	Operational earnings efficiency — mean 40.08%
Profit_Category	4 buckets: High (40%+), Medium, Low, Very Low	Margin segmentation for cross-analysis

Column	Formula	Business Purpose
Risk_Level	4 buckets from Fraud_Risk_Score: High/Med/Low/Very Low	Risk categorisation for portfolio analysis
Is_High_Performer	Performance_Score $\geq$ 90 $\rightarrow$ 1 else 0	Binary flag — 71.3% of transactions flagged

05 TOOLS & TECHNOLOGIES USED

Tool	Phase	Purpose
Python 3.11	Phase 1	EDA, 6 negative value fixes, 7 derived columns, clean CSV export
pandas / numpy	Phase 1	Data manipulation — abs(), apply(), cut(), astype()
Microsoft Excel	Phase 2	Branch KPI PivotTable, conditional formatting, Status cross-tab, KPI Summary
MySQL 8.0	Phase 3	CREATE TABLE, LOAD DATA INFILE, 6 indexes, 7 queries, 1 VIEW
Power BI Desktop	Phase 4	MySQL connection, 12 DAX measures, 3-page 24-visual dashboard
DAX	Phase 4	Financial measures: revenue, profit, margin, ROI, compliance
JSON Theme	Phase 4	Corporate Finance Gold & Dark — dark navy canvas, gold accents

06 PHASE 2 — EXCEL: FINANCIAL MODEL & KPI ANALYSIS

- Branch KPI PivotTable: PivotTable on 'Branch KPIs' sheet with Branch_ID as rows and 7 KPI measures (Avg Revenue, Avg Net Margin %, Avg ROI %, Avg Expense Ratio %, Avg Gross Margin %, Avg Performance, Avg Compliance). Conditional formatting: Color Scale Green $\rightarrow$ Red on Net Margin %, Data Bars on ROI %, Color Scale Red $\rightarrow$ Green on Expense Ratio %, Traffic Light icons on Performance Score.
- Financial Status Cross-Tab: Two PivotTables on 'Status Summary' sheet — Region $\times$ Financial_Status count and Department $\times$ Financial_Status count — both with Color Scale formatting revealing that West region has most 'Poor' transactions (59) and North has most 'Moderate' transactions (141).
- KPI Summary Sheet: Formula-driven KPI reference table with 10 metrics linking to the raw data sheet — Total Revenue (\$4.67B), Total Net Profit (\$1.36B), Total EBITDA (\$1.14B), Avg Net Margin %, Avg ROI %, Avg Gross Margin %, Avg Compliance Score, High Performers %. Conditional formatting applied to each KPI value cell (green/amber/red thresholds).

07 PHASE 3 — MYSQL: IMPORT, INDEXES & QUERIES

Import Summary

- Imported financial_clean.csv (Python output, 8,900 rows, 46 columns) via LOAD DATA INFILE — completed in seconds with 0 warnings. Transaction_ID used as VARCHAR PRIMARY KEY.

- 6 performance indexes created on Region, Department, Branch_ID, Financial_Status, Profit_Category, Risk_Level — enabling all analytical queries to run efficiently without full table scans.

7 Analytical Queries + Results

Query	Purpose	Key Results
Q1 — Overall KPIs	Revenue, profit, costs, EBITDA, margins, ROI, performance	\$4.67B revenue · \$1.36B net profit · \$2.29B operating cost · \$1.14B EBITDA · 48.01% avg margin · 37.86% gross margin · 18.56% ROI
Q2 — Financial Status	Distribution, avg revenue, margin, performance per status	Excellent 77.07% (perf 96.63) · Good 13.20% (perf 78.00) · Moderate 6.82% (perf 63.51) · Poor 2.91% (perf 46.52, margin 51.27)
Q3 — By Region	Revenue, profit, margin, ROI, compliance per region	South #1 revenue \$954.8M · East #1 margin 49.20% · North lowest margin 48.69% · all regions within 2.6% margin range
Q4 — By Department	Revenue, margin, expense ratio, ROI, fraud risk per dept	Investment #1 margin 50.49% · Audit #1 revenue \$801.4M · Treasury lowest margin 45.67% · Operations lowest expense ratio 77.72%
Q5 — Branch Rankings	Top 10 and Bottom 10 branches by avg net margin %	Top: Branch 1025 (64.33%) · Branch 1005 (62.20%) · Bottom: Branch 1017 (36.32%) · Branch 1077 (38.14%) — 28pp margin gap
Q6 — Profit×Risk Cross	Profit category × Risk level combinations: count, revenue, margin, compliance	High profit + any risk level avg margin ~99% · Low profit categories avg 17% regardless of risk level — risk level does NOT significantly affect margin
Q7 — VIEW	vw_financial_summary — pre-aggregated by 5 dimensions for Power BI	Used in Power BI for matrix heatmap and risk/profit distribution without real-time aggregation

08 PHASE 4 — POWER BI: DAX & DASHBOARD

12 DAX Measures

Measure	Result
Total Transactions	8,900
Total Revenue	\$4,667,647,698.83
Total Net Profit	\$1,364,116,558.93
Total Operating Cost	\$2,293,341,164.45
Total EBITDA	\$1,139,068,843.45
Avg Net Margin %	48.01%
Avg ROI %	18.56%
Avg Gross Margin %	37.86%
Avg Compliance Score	80.03

Measure	Result
High Performer Count	6,344 (71.3%)
Excellent Status Count	6,859 (77.07%)
Poor Status Count	259 (2.91%)

3-Page Dashboard — 24 Visuals

Page	#	Visual	Key Insight
P1 Financial Overview	1–6	6 KPI Cards	\$4.67B · \$1.36B · \$2.29B · \$1.14B · 48.01% · 18.56%
P1 Financial Overview	7	Waterfall Chart	Revenue by Financial Status — Excellent dominates
P1 Financial Overview	8	Bar Chart	Total Revenue by Region — South #1 at \$954.8M
P1 Financial Overview	9	Donut Chart	Financial Status — Excellent 77.07%
P1 Financial Overview	10–11	Slicers	Financial_Status · Region
P2 Dept & Profitability	12	Bar Chart	Avg Net Margin by Dept — Investment leads 50.49%
P2 Dept & Profitability	13	Column Chart	Revenue vs Net Profit by Department
P2 Dept & Profitability	14	Bar Chart	Profit Category — High (3,145) dominates
P2 Dept & Profitability	15	Matrix Heatmap	Region × Department Net Margin — color-coded grid
P2 Dept & Profitability	16	Donut Chart	Risk Level — roughly equal 4-way split (~25% each)
P2 Dept & Profitability	17–18	Slicers	Department · Profit_Category
P3 Branch & Risk	19	Bar Chart	Top 10 Branches — Branch 1025 leads at 64.33%
P3 Branch & Risk	20	Bar Chart	Bottom 10 Branches — Branch 1017 worst at 36.32%
P3 Branch & Risk	21	Column Chart	Avg ROI % by Region — South #1 at 19.04%
P3 Branch & Risk	22	Scatter Chart	Compliance vs Performance by Financial_Status
P3 Branch & Risk	23–24	Slicers	Risk_Level · Financial_Status (synced)

09 KEY INSIGHTS & RESULTS

A. Overall Financial Portfolio

- The \$4.67B revenue portfolio generates \$1.36B net profit (48.01% avg margin) after \$2.29B operating costs — but the 81.02% average expense ratio signals that for every \$1 of revenue, \$0.81 goes to operating costs before any other deductions. The strong EBITDA margin (40.08%) indicates solid operational earnings despite high expense ratios.
- 71.3% of all 8,900 transactions are flagged as High Performers (Performance Score ≥ 90) and 77.07% carry Excellent Financial Status — confirming the overall portfolio quality is strong, but the 2.91% Poor-status transactions (259 records) and 6.82% Moderate-status transactions (607 records) represent priority intervention candidates.

B. Regional Performance

- South region leads by total revenue (\$954.8M, 1,814 transactions) and ROI (19.04%) while East leads by net profit margin (49.20%, 1,709 transactions). The 2.6pp margin range (East 49.20% to North 48.69%) is surprisingly narrow — confirming highly standardised IoT financial processes across geographic regions with minimal regional operational advantage.
- East region has the fewest transactions (1,709 vs South 1,814) but the highest margin, suggesting fewer but higher-quality transactions. West region has the most Poor-status transactions (59) — a priority region for quality improvement.

C. Department Performance

- Investment department leads by net profit margin (50.49%) while Audit leads by total revenue (\$801.4M). Treasury has the lowest margin (45.67%) and highest expense ratio (80.46%), suggesting cost optimisation opportunity. Operations has the lowest expense ratio (77.72%) — the most cost-efficient department despite ranking 5th by revenue.
- Fraud risk scores are essentially identical across all 6 departments (0.495–0.511 avg) — confirming fraud risk is not department-specific but distributed uniformly across the IoT transaction portfolio, likely reflecting systematic IoT security characteristics rather than department-level vulnerability.

D. Branch Performance Gap

- Branch 1025 leads all 100 branches with 64.33% average net margin (100 transactions, ROI 19.56%, Performance 90.66%). Branch 1017 ranks last at 36.32% — a 28.01 percentage point gap between best and worst performing branches. This 28pp spread represents significant variance in branch-level financial efficiency that warrants management investigation.
- Bottom branches (1017 at 36.32%, 1077 at 38.14%) maintain performance scores above 89 despite low margins — suggesting operational performance is not the root cause of low profitability; rather, transaction mix, cost allocation, or regional pricing factors likely explain the margin gap.

E. Risk-Profit Cross Analysis

- High-profit transactions (40%+ margin) maintain ~100% average net margin across ALL risk levels (High Risk 100.43%, Low Risk 101.43%, Medium Risk 98.68%, Very Low Risk 98.74%) — confirming that for this IoT financial dataset, profit category and risk level are INDEPENDENT variables. Higher margins do not attract disproportionate fraud/credit risk.
- Low-profit transactions average ~17% margin regardless of risk level — confirming margin is determined by transaction economics, not risk exposure. This is a key insight for risk management: high-margin transactions can be pursued without accepting higher fraud risk trade-offs.

F. Complete KPI Summary

KPI	Value	KPI	Value
Total Revenue	\$4,667,647,698	Total Net Profit	\$1,364,116,559
Total Operating Cost	\$2,293,341,164	Total EBITDA	\$1,139,068,843
Avg Net Margin %	48.01%	Avg Gross Margin %	37.86%
Avg ROI %	18.56%	Avg Expense Ratio %	81.02%
Avg Performance Score	90.45	Avg Compliance Score	80.03
Excellent Status	6,859 (77.07%)	Poor Status	259 (2.91%)
High Performers	6,344 (71.3%)	Total Transactions	8,900
Top Region (Revenue)	South — \$954.8M	Top Region (Margin)	East — 49.20%
Top Department	Investment — 50.49%	Lowest Department	Treasury — 45.67%
Top Branch	1025 — 64.33% margin	Bottom Branch	1017 — 36.32% margin
Branches Analyzed	100	Branch Margin Gap	28.01 percentage points

10 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Dataset Does Not Match Planned Workflow Description

- Problem: The project brief described 'multi-company data with revenue, expenses, profit margins, and growth metrics' with planned YoY growth, CAGR, and company comparison analysis. The actual dataset is an IoT transaction-level file with no Company column, no Date/Year column, and no multi-company structure — making YoY growth, CAGR calculations, and company comparisons literally impossible.
- Solution: Performed thorough EDA before writing any instructions — immediately identified the structural mismatch. Redesigned the workflow to use Region (5 values) as the 'business unit' equivalent, Branch_ID (100 values) as the 'company' equivalent for ranking analysis, and Financial_Status/Profit_Category as performance segmentation dimensions. All originally planned analyses were adapted to the actual data structure rather than forcing incompatible calculations.

Challenge 2 — 6 Negative Sensor Values in Physical Measurement Columns

- Problem: Network_Latency_ms (3 negative values, min=-1.37) and Transaction_Processing_Time_ms (3 negative values, min=-6.59) contained physically impossible negative measurements — latency and processing time cannot be negative. These were sensor calibration artifacts.
- Solution: Applied abs() (absolute value) function to both columns in Python — converting -1.37 → 1.37 and -6.59 → 6.59. Used absolute value rather than setting to 0 or removing rows — preserving the magnitude information while correcting the impossible sign. Verified both columns showed 0 negatives after fix.

Challenge 3 — Extremely High Expense Ratio Anomalies

- Problem: Expense_Ratio_Pct ($\text{Operating_Cost}/\text{Revenue} \times 100$) showed a mean of 81.02% which seemed reasonable, but max of 950% and some values above 100% — meaning some transactions have Operating_Cost exceeding Revenue by 9.5×. These extreme values are potentially data quality issues.
- Solution: Kept all values as-is after investigation — in IoT financial data, some transactions can legitimately have high operating costs relative to transaction revenue (particularly initial device setup, high-maintenance operations, or loss-leading transactions). The overall portfolio health (48% net margin) confirms these outliers don't dominate. Noting them as valid business data rather than errors.

Challenge 4 — Waterfall Chart Configuration in Power BI

- Problem: Power BI's native Waterfall chart requires specific field configuration — Category drives the breakdown bars, Y-axis drives the bar heights, and Breakdown creates the color-coded increase/decrease segments. Getting the Financial_Status waterfall to display meaningful revenue progression required understanding these field mappings.
- Solution: Used Financial_Status as Category (X-axis), Total Revenue as Y-axis, and left Breakdown empty for a simple stepwise display. This produced the Excellent→Good→Moderate→Poor revenue waterfall showing how total revenue is distributed across status categories — not a traditional profit waterfall but a valid revenue breakdown visualization using the available data dimensions.

Challenge 5 — Branch Bar Charts Showing All 100 Branches (Unreadable)

- Problem: Creating bar charts for all 100 Branch_IDs produced an unreadable chart with 100 bars cramped into the visual, defeating the purpose of identifying top/bottom performers.
- Solution: Added Top N filter in the Filters pane: Filters on this visual → drag Branch_ID → Filter type: Top N → Top 10 / Bottom 10 → by Avg Net Margin %. This automatically shows only the highest/lowest 10 branches and updates dynamically when slicers are applied — creating two focused visuals that answer the business question directly.

11 SKILLS LEARNED & TAKEAWAYS

- Workflow Adaptation from EDA: Recognising when actual data structure doesn't match the project brief — and redesigning the entire workflow to fit the real data rather than forcing incompatible analyses on it. Using Region/Branch as analytical equivalents for Company/Year.
- Financial Ratio Engineering in Python: Calculating 7 financial ratios (Net Margin, Expense Ratio, Gross Profit, EBITDA Margin, Profit Category buckets, Risk Level buckets, High Performer flag) — understanding that financial analysis requires pre-computing ratios before any SQL or BI work.
- Excel as Analytical Middle Layer: Using Excel PivotTables with conditional formatting as a standalone analytical artifact (branch KPI comparison, cross-tabulation, formula-driven KPI summary) — understanding Excel's role in the 4-tool pipeline as an exploratory and presentation layer.
- Waterfall Chart Design: Building Power BI's native waterfall chart with financial status breakdown — understanding Category, Y-axis, and Breakdown field roles and how to configure it for non-traditional uses (revenue distribution vs profit cascade).
- Top N Filter in Power BI: Using Filters pane → Top N filter type to automatically limit branch bar charts to 10 rows — a professional dashboard technique that avoids unreadable 100-bar visuals while remaining dynamic with slicers.

- Risk-Profit Independence Analysis: Discovering through Q6 cross-analysis that profit category and risk level are statistically independent in this dataset — a non-obvious and actionable insight that required designing a CROSS JOIN-style GROUP BY query to reveal.

12 CONCLUSION

- Processed 8,900 IoT financial transaction records through a complete 4-tool Advanced pipeline — Python EDA fixed 6 negative sensor values and engineered 7 financial derived columns, Excel built a branch KPI comparison model with conditional formatting thresholds, MySQL imported the clean 46-column dataset and executed 7 analytical queries revealing \$4.67B total revenue and 48.01% average net margin.
- Identified East as highest-margin region (49.20%), Investment as highest-margin department (50.49%), Branch 1025 as top performer (64.33% margin vs Branch 1017's 36.32% worst — a 28pp gap), and a critical risk-profit independence finding: high-margin transactions carry NO elevated fraud risk compared to low-margin transactions in this IoT portfolio.
- Built a 3-page Power BI dashboard (24 visuals, 12 DAX measures) covering Financial Overview, Department & Profitability, and Branch Performance & Risk — with a Region×Department matrix heatmap, compliance vs performance scatter, top/bottom branch rankings using Top N filters, and Sync Slicers across all pages.
- Demonstrated Advanced portfolio capability: workflow redesign based on EDA evidence, multi-tool pipeline execution (Python + Excel + MySQL + Power BI), financial ratio engineering, cross-dimensional risk analysis, and executive-quality dashboard design with a Corporate Finance Gold & Dark theme. Project 8 of 9 — second Advanced project completed.